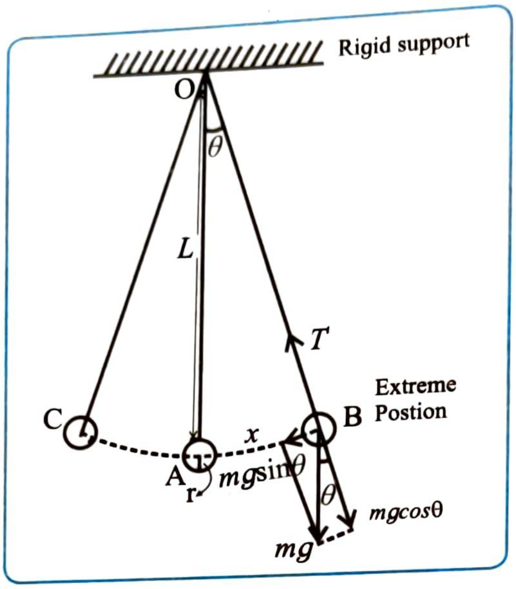
ACTIVITY NO. 2
SECOND'S PENDULUM

Aim : To determine the length of the seconds pendulum

Apparatus : Metal sphere (bob), inextensible string, retort stand, meter scale, stop watch, vernier calliper

Diagram :



Formula :

$$\frac{L_S}{T_S^2} = \frac{L_S}{4} = \left(\frac{L}{T^2}\right)_{mean} \qquad \therefore L_S = 4\left(\frac{L}{T^2}\right)_{mean}$$

Procedure :

1. Suspend the bob (radius given) from the fixed support of a retort stand by means of inextensible string
2. Adjust the length of the string to 80 cm.
3. Displace the bob to its equilibrium position and release, so that it performs linear S.H.M.
4. Measure the time (t) for 20 oscillations.
5. Record the observation for two sets of oscillations..
6. Hence find the time period T = t/20 (seconds).
7. Repeat the steps 4 to 7 by adjusting the lengths of the string 90 and 100 cm.

Radius of the bob = ..0.5... cm

Period of second's pendulum $T_S = 2$ sec

Observations for period oscillations T : Period

Obs. No.	Length of the string ℓ cm	Length of the pendulum L=(ℓ + r) cm	Time for 20 oscillations t_1 sec	t_2 sec	Mean t(sec)	T = t/20 sec	$\frac{L}{T^2}$ $\frac{cm}{S^2}$
1	80	80.5	40	42	41	2.05	19.155
2	90	90.5	38	40	39	1.95	23.800
3	100	100.5	38	36	37	1.85	29.364

mean = $\frac{L}{T^2}$ 24.106 cm s^{-2}

Calculations :

(1) $\frac{L}{T_1^2} = \frac{30.5}{4.2025}$
$= 19.155$

(2) $\frac{L}{T_2^2} = \frac{90.5}{3.8025}$
$= 23.800$

(3) $\frac{L}{T_3^2} = \frac{100.5}{3.4225}$
$= 29.364$

$$L_S = 4 \times \left(\frac{L}{T^2}\right)_{mean} = 4 \times 24.106 = 96.42 \text{ cm}$$

$\Rightarrow \frac{L_S}{T_S^2} = (24.106)$

Result :

Length of seconds pendulum T_s = 2 sec.

$\frac{96.42}{(24.106)} = T_S^2$

$4.000 = T_S^2$

$2 = T_S$

Precautions :

1. Amplitude of oscillations should be small
2. String used in the experiment should be inextensible and light.

Questions

1. Define Amplitude in SHM

The maximum displacement of a particle executing SHM from its mean position.

2. State the nature of vibrations of a pendulum (LSHM)

For small angular displacements, the motion of the bob of a simple pendulum is linear simple harmonic motion about its mean position.

3. Define force constant

The restoring force per unit displacement produced in a spring is called its force constant (spring constant).

4. What is a seconds pendulum ?

A simple pendulum whose time period is 2 seconds is called a seconds pendulum.

Remark and sign of teacher :